

Lecture 8: Relational Database Design Theory

Amit Kumar Dhar

August 20, 2025

Introduction to Design Theory

Why is Design Theory Needed?

Good database design is crucial to avoid data management problems. Without a solid theoretical foundation, we can run into issues like:

- **Data Redundancy:** The same information is stored multiple times, wasting space and risking inconsistency.
- **Insertion Anomaly:** Inability to add a new record because some data is missing.
- **Update Anomaly:** Changing information in one place requires changing it in many other places.
- **Deletion Anomaly:** Deleting one piece of information unintentionally deletes other, unrelated information.

Goal: Design a schema that is "good" by formally defining properties that prevent these anomalies. The main theories are based on **Functional Dependencies** and **Normal Forms**.

Functional Dependencies

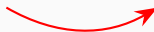
Functional Dependencies (FDs)

A functional dependency is a constraint between two sets of attributes in a relation.

Definition: An FD is denoted as $\alpha \rightarrow \beta$, where α and β are sets of attributes. This means that the values of the attributes in α uniquely determine the values of the attributes in β .

Example: Consider a 'Students' relation.

sid	sname	major
101	Alice	CS
102	Bob	EE
101	Alice	CS


determines

When does an FD Hold?

An FD $\alpha \rightarrow \beta$ **holds** on a relation if, for any two tuples, whenever they have the same values for all attributes in α , they must also have the same values for all attributes in β .

Example 1: FD that HOLDS

- Consider $\text{sid} \rightarrow \text{major}$. For any two rows with the same 'sid', the 'major' is also the same. This holds.

Example 2: FD that DOES NOT HOLD

- Consider $\text{major} \rightarrow \text{sname}$ in our 'Students' table.

sid	sname	major
101	Alice	CS
102	Bob	EE
103	Charlie	CS

Keys and Functional Dependencies

FDs provide a formal way to define keys:

- **Superkey:** A set of attributes α is a superkey for a relation R if $\alpha \rightarrow R$ holds. That is, α functionally determines all attributes in R .
 - Example: $\{\text{sid}, \text{sname}\}$ is a superkey for Students.
- **Candidate Key:** A candidate key is a *minimal* superkey. No proper subset of a candidate key is also a superkey.
 - Example: $\{\text{sid}\}$ is a candidate key for Students.
- **Primary Key:** One of the candidate keys chosen by the database designer to be the main identifier for tuples.

Reasoning About FDs

Follows and Equivalence

- **Follows:** An FD f **follows** from a set of FDs F if every relation that satisfies all FDs in F also satisfies f . The set of all FDs that follow from F is called the **closure** of F , denoted F^+ .
- **Equivalence:** Two sets of FDs, F and G , are **equivalent** if $F^+ = G^+$. That is, they logically imply the exact same set of dependencies.

Example: Given $F = \{A \rightarrow B, B \rightarrow C\}$, the FD $A \rightarrow C$ *follows* from F .

Rules for Reasoning: Armstrong's Axioms

To reason about FDs, we use a set of inference rules called Armstrong's Axioms.

- **Reflexivity (Trivial FD):** If $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$.
 - Example: $\{\text{sid}, \text{sname}\} \rightarrow \text{sname}$ is a trivial FD.
- **Augmentation:** If $\alpha \rightarrow \beta$, then $\gamma\alpha \rightarrow \gamma\beta$.
 - Example: If $\text{sid} \rightarrow \text{sname}$, then $\{\text{sid}, \text{major}\} \rightarrow \{\text{sname}, \text{major}\}$.
- **Transitivity:** If $\alpha \rightarrow \beta$ and $\beta \rightarrow \gamma$, then $\alpha \rightarrow \gamma$.
 - Example: If $\text{sid} \rightarrow \text{major}$ and $\text{major} \rightarrow \text{dept_head}$, then $\text{sid} \rightarrow \text{dept_head}$.

From these, we can derive the **Splitting/Combining (Union)**

Rule: $\alpha \rightarrow \beta$ and $\alpha \rightarrow \gamma$ is equivalent to $\alpha \rightarrow \beta\gamma$.

Attribute Closure

Computing the Closure of Attributes

The **closure of a set of attributes** α with respect to a set of FDs F , denoted α^+ , is the set of all attributes that are functionally determined by α .

Algorithm to compute α^+ :

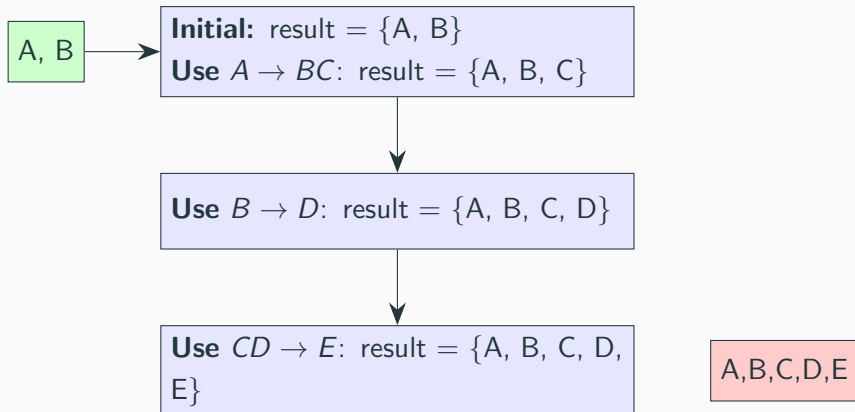
1. Initialize `result` = α .
2. Repeatedly iterate through the FDs in F :
3. If there is an FD $\beta \rightarrow \gamma$ in F such that $\beta \subseteq \text{result}$, then add all attributes in γ to `result`.
4. Repeat until `result` no longer changes.

Example: Computing Attribute Closure

Let $R = (A, B, C, D, E)$ and

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$.

Let's compute $\{A, B\}^+$.



Since $\{A, B\}^+$ contains all attributes of R , $\{A, B\}$ is a superkey.

Thank You!
Questions?